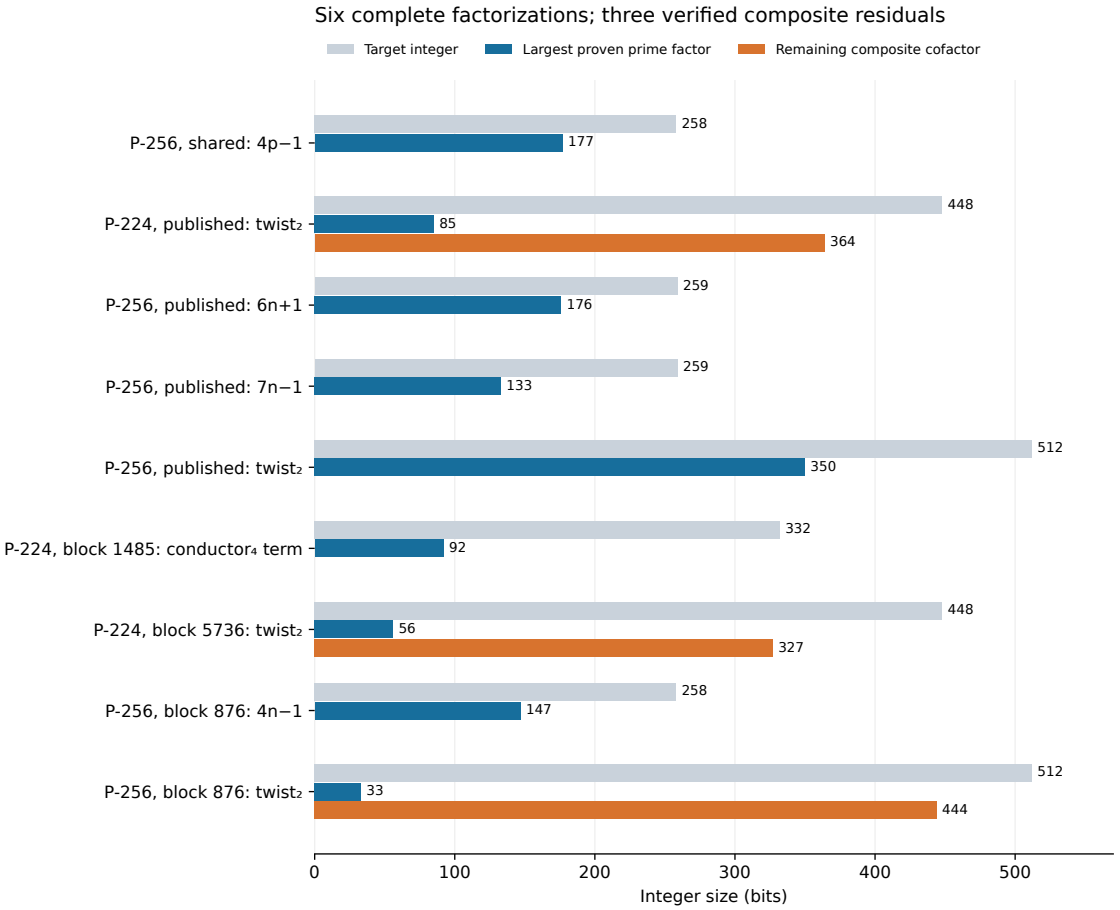
Continuing the NIST curve factorizations

Run `nist-factor-continuation-20261021T1424Z` · 21 October 2026

Six of nine distinct integers are fully factored into proven primes. Three retain composite cofactors. This resolves **eleven of fourteen** timeout-valued fields across seven of the ten affected rows in the earlier 438-case upstream DiSSECT run. The original outputs are preserved.

The fixed panel comprises P-224 stream blocks 777, 1485 and 5736, P-256 block 876, and published P-224/P-256 controls. Imported fixture keys and exact hex parameters identify the models. No model with the opposite sign of `b` was substituted.

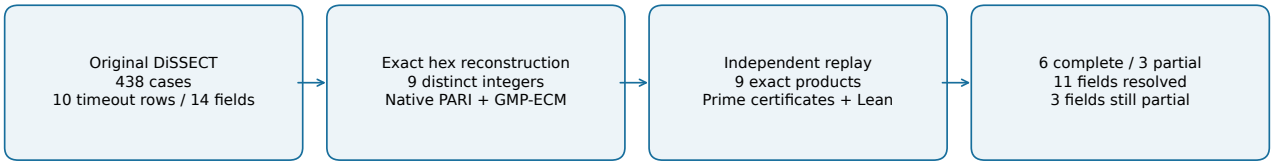


Run nist-factor-continuation-20261021T1424Z · nine fixed integers, one partition each.
Exact bit lengths from results.json; no sampling uncertainty. A composite cofactor is not a prime factor.

The graph plots exact bit lengths serialized by the native verifier: one partition per fixed integer, with no sampling interval. A residual cofactor is explicitly composite. It is not a proven prime factor. Full decimals: factorizations and cofactors.

Both P-256 `4p-1` square-part roots and the skipped P-256 `4n-1` root are 1. Resolved largest-factor sizes are 147 bits for skipped `4n-1`, 176 for published `6n+1`, and 133 for published `7n-1`. These facts do not explain the historical skips.

Inputs, executed search and controls



Different denominators: cases → timeout rows/fields → distinct arithmetic targets.
This supplement preserves original timeouts; it does not establish a historical skip reason.

Let `t=p+1-n`. Native PARI reconstructs the degree-2 extension-twist orders `(p-1)^2+t^2`, the degree-4 conductor-ratio factorization target `abs(t*(t^2-2*p))`, and the required `kn±1` and `4p-1` integers directly from hex. The conductor-ratio term is not, by itself, an endomorphism-ring conductor. Despite its name, the upstream `square_4p1` trait uses minus one for both inputs.

Two P-256 `4p-1` requests share an integer; skipped `4n-1` duplicates another request. Three bit-length timeout fields duplicate their factorization fields. Thus ten rows mean fourteen fields and nine integers. The additive erratum preserves the initial protocol's eleven-field miscount.

PARI/GP 2.17.2 used trial division, exact algebraic partitioning, read-only FactorDB hints and 600-second native factorization attempts. All hints were checked by exact division. Six targets completed in this stage. GMP-ECM 7.0.6 then produced more factors for each remaining target lineage.

Target lineage / stage	B1	Curves tried	Stage 2 done
Published P-224 / first split	100000	24	24
Published P-224 / remainder	100000	76	76
P-224 block 5736 / first split	100000	6	6
P-224 block 5736 / next split	100000	4	4
P-224 block 5736 / remainder	100000	90	90
P-256 block 876 / first split	100000	1	0
P-256 block 876 / remainder	100000	99	99
Each of three residuals	1000000	200 each	200 each
Each of three residuals	3000000	100 each	100 each

After a split, only the unused curve allocation continued on its smaller cofactor. The final 300 curves found no new factor. Every curve parameter, saved residue, command and exit is retained; the attempt inventory enumerates all 15 ECM subprocesses including controls. Nonzero even exits can encode factors.

The first random positive control reported its entire input, a trivial result, and remains a failed control. The corrected deterministic control at sigma `1:1184114830` finds `1000003` with cofactor `1000033`. The wrong-target replay control rejects its input.

At most two workers ran together. Each ECM cofactor attempt had a 600-second limit and 1024 MB stage-2 memory cap. The unisolated Linux 6.18.55 x86-64 container had a four-CPU, 16 GiB cgroup limit. No remote machines ran. This is a completion search, without performance or speedup claims.

Verification, failures and remaining composites

Separate native replay reconstructs expected targets, validates every saved PARI prime certificate using `primecertisvalid`, separately calls `isprime`, checks positive exponents, composite residuals and exact products. All nine partitions pass. It is a separate process using the same PARI arithmetic implementation, not an independent implementation.

Lean 4.35.0 proves all nine exact partition products over natural numbers with `by decide`, including the three explicit composite cofactors. The repository gate audits nine theorems, zero failing and zero trusted native proofs; every axiom list is empty. **Lean proves products, not primality.** Native PARI certificates establish the separate primality claims. `lean4checker` was unavailable. The proof map names every theorem and identity record.

The first PARI wrapper falsely rejected a harmless stack warning. Three native searches hit their 600-second alarms and retained partial partitions. One certificate-generation attempt overflowed the default 8 MB stack; the fresh 128 MB initial / 2 GB maximum-stack attempt passed. A Lean 4.25 build proved five products but failed in the incompatible axiom auditor; 4.35.0 passes. The initial constant-expression encoding exceeded the JSON nesting limit; the corrected native decimal-chunk encoding passes separate issue/check invocations. Failed evidence is preserved.

The required library suite reported 4465 passed, 9 failed and 119 ignored when loopback networking was denied. All nine failures were in `pollard_collab`. Rerunning that whole module with network permission passed 44/44 tests, covering every failure. This is a full-suite run plus a targeted retry, not a newly observed single all-green full-suite run. Shell syntax, identity-record checks and all nine factor replays pass.

Unresolved cofactors

Each decimal below is composite. Spaces/line breaks are for layout only; the exact strings and proven prime factors are in results.json.

P-224 stream block 5736 — 327 bits

```
19106509893925638590876479307257909680692826
55793904142972730384691966042890337782660059
78480168453
```

Published P-224 — 364 bits

```
26824805404542206456882676187978690075366455
86954534173696932703925808640719391831005646
0278604794052610896253
```

P-256 stream block 876 — 444 bits

```
33492266050798185653682380062483799213318454
80165396349313406472105973593694452090729593
14966144558580760484371844990904784202048883
53
```

No smallest-prime-factor lower bound follows from exhausted ECM allocations. Complete prime-certified refinements would close this factorization gap. The forty original generator-dependent null cases and upstream torsion and degree-2 trace defects remain separate. No historical skip reason or new discrete-logarithm attack is established.

Replay and evidence sources

From the study's `factorization/` directory, with PARI/GP 2.17.2:

```
sha256sum -c runs.tar.xz.sha256
tar -xJf runs.tar.xz
bash replay_results.sh runs /tmp/nist-factor-replay-01
```

Choose a fresh replay directory. Set `GP_BIN` to a single executable path if needed. New JSON encodes large integers as decimal strings. Do not round-trip them through JSON numbers.

The deterministic archive contains all new attempts, failures, parameters, residues, raw hint responses, intermediate partitions, final certificates and verification logs. Original absolute paths record provenance; the replay builds fresh descriptors for the extracted location. Repacking instructions and native identity-check build commands are in the study README.

The unchanged external `inputs/upstream-dissect.zip` has 932 files: source, 438 cases, logs and audit. Pinned upstream revision: `8edc3030fb91411a55aa6c07d050507d840c5e55`. The fixture matches the external bundle exactly on the six models' `p,a,b,n,h`. Input and executable hashes are in provenance.json. The manual workflow was not dispatched. Conductor was unavailable; the documented fallback was used.

Formal product map

Target	Identity certificate
Shared P-256 $4p-1$	`IDC1h31218b677856b8b9`
Published P-224 twist_2	`IDC1hdb2af39b925576a1`
Published P-256 $6n+1$	`IDC1hdecb8fce71e2eab7`
Published P-256 $7n-1$	`IDC1h512190b8fcf5d7fe`
Published P-256 twist_2	`IDC1h969e7da713aca14e`
P-224 block 1485 conductor_4 term	`IDC1h7c5ecb55bf5e2915`
P-224 block 5736 twist_2	`IDC1he5e12c8967667144`
P-256 block 876 $4n-1$	`IDC1h0ba14f42c133d8da`
P-256 block 876 twist_2	`IDC1hb6ab9163ca2cf443`

The constant-identity records evaluate over `2^61-1` and link the exact Lean source digest. Modular equality alone would not prove integer equality; the linked Lean kernel proofs establish the exact products.

Sources and scope

- Anthropic: ECDSA curve origins
- Filippo Valsorda: Seeds solved
- Pinned DiSSECT source: square_4p1, twist_order, conductor and kn_factorization
- PARI prime certificates
- GMP-ECM
- PR 2379: jobs and continuation evidence

No existing canonical graph tracks these factorization entries. IC scoreboards/timeline, curve/performance figures and seed-study visuals remain unchanged. The new graph and rendered copies use only recorded native values; plotting Python performs layout. The remaining composite-cofactor problem is linked from the study, together with exact acceptance conditions.